

# Clause Reviewer

---

## AI Contract Review — Business Case

IEOR 4576 | Yiming Zhu | May 2026

### 1. User

---

Clause is built for the independent professional who signs contracts regularly but cannot justify the cost of a lawyer for routine review.

The primary user is a freelancer, independent consultant, or small agency owner — a graphic designer, software developer, marketing consultant, or copywriter — who signs 5 to 20 contracts per year. They are typically earning between \$60,000 and \$200,000 annually and operate in a market where the client almost always provides the contract template. They have legal exposure but not legal resources.

### 2. Problem

---

There are over 70 million independent workers in the United States. The vast majority sign contracts they do not fully understand. The cost of getting it wrong is real: a broad non-compete can lock a designer out of their industry for two years; a full IP assignment clause with no portfolio carve-out strips a developer of the work they need to win their next client.

Today, users have three options — all of which fail them:

- (1) Skip review entirely. The most common choice. Users skim, miss the dangerous language buried in it, and sign.
- (2) Ask a lawyer. A one-hour contract review at a mid-market law firm costs \$300 to \$500. For a \$3,000 freelance project, that is a 10 to 17 percent overhead. Most users do not do this.
- (3) Use a general-purpose LLM. ChatGPT and similar tools will summarize a contract but do not produce structured, graded, actionable analysis. They cannot precisely distinguish a mutual liability cap (fair) from a one-sided uncapped indemnity (high risk). They do not cite verbatim clause text. They do not generate replacement language.

Clause addresses this gap: structured, calibrated, contract-type-aware review in under 30 seconds, at a cost that is a relatively small when compared to professional legal fees.

### 3. Economics & Operations

---

#### Pricing Model

Subscription with a usage tier:

- (1) Free tier: 3 contract per month (acquisition)
- (2) Pro: \$15/month — 20 contracts included, \$3.99 per additional
- (3) Business: \$79/month — 150 contracts, multi-user, exportable reports

#### Unit Economics — One Pro User, One Month

| Metric       | Value            | Notes                                     |
|--------------|------------------|-------------------------------------------|
| Revenue      | \$15.00          | Monthly subscription, Pro tier            |
| Total COGS   | ~\$1.60          | 20 analyses × \$0.08 each (model + infra) |
| Gross margin | ~\$13.40 (89.3%) | Before S&M and R&D                        |

## Token Economics — Per Analysis

Each contract analysis triggers four Gemini 2.5 Flash calls (Risk, Gap, Negotiation, Critic) plus embedding calls for ingestion. A typical 10-page contract generates approximately:

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
| <b>Ingestion (embedding)</b> | ~2,500 tokens — chunking + Vertex AI text-embedding-004        |
| <b>Classification call</b>   | ~600 tokens input / 256 output — doc type detection            |
| <b>Risk agent</b>            | ~4,000 tokens input / 2,000 output — full text analysis        |
| <b>Gap agent</b>             | ~4,000 tokens input / 2,000 output — full text analysis        |
| <b>Negotiation agent</b>     | ~4,000 tokens input / 4,000 output — suggestions with rewrites |
| <b>Critic agent</b>          | ~1,200 tokens input / 1,024 output — verdict synthesis         |
| <b>Total per analysis</b>    | ~15,800 tokens input / ~9,280 output                           |

At Gemini 2.5 Flash pricing (approximately \$0.075 per million input tokens, \$0.30 per million output tokens), one full analysis costs under \$0.01 in model API fees. The dominant cost is the parallel Gemini calls, not embedding. At \$0.08 total per analysis including infra, a Pro user running 20 contracts per month generates \$1.60 in COGS against \$15 in revenue — an 89.3% gross margin.

## Where It Breaks

- (1) Very long contracts (50+ pages) increase token costs proportionally. Mitigation: truncate to 4,000 words per agent call, which covers the vast majority of freelance and commercial contracts.
- (2) High-volume business tier users running 20+ analyses per month compress margin. At 150 analyses × \$0.08 = \$12.00 COGS against \$79 revenue, gross margin remains above 84%.
- (3) Customer acquisition cost is not modelled here but is the primary scaling risk. The product is well-suited to organic acquisition through professional networks, communities, and SEO on contract-review queries.
- (4) Legal trust and liability are the largest non-token constraints. Clause is positioned as first-pass contract risk screening, not legal advice. It only identifies risky clauses, explains why they matter, suggests negotiation language, and recommends escalation to counsel for high-severity issues.

## Break-even

Fixed monthly infra (Cloud Run, domain, monitoring) is approximately \$40. Break-even occurs at 4 paying Pro users. At 100 Pro users, monthly gross profit is approximately \$1,340 before S&M and R&D salaries.

## 4. Why These Technical Choices

### Parallel multi-agent

Contract review naturally decomposes into independent, parallel workstreams: risk analysis, gap detection, and negotiation suggestions do not depend on each other's results. Running them sequentially would take 45 seconds per analysis — too slow for a real-time product. Running them in parallel with `asyncio.gather` reduces this to 12-18 seconds. This is a direct latency-to-retention business requirement: users will abandon an analysis that takes

longer than 20 seconds.

### **Generator-Critic pattern**

The three specialist agents can produce inconsistent or miscalibrated findings independently. The Critic agent reads all three outputs and synthesises a single calibrated verdict with exactly three action items. This addresses a product requirement: users need one clear answer, not three separate reports to reconcile.

### **Full-text analysis instead of RAG retrieval**

The initial design used RAG to retrieve relevant contract fragments before analysis. Testing revealed that fragment-based retrieval introduced hallucination: the model received a clause without surrounding context and completed it from training data rather than the actual contract. Switching to full-text analysis (the entire contract passed to each agent) eliminated this. The RAG infrastructure remains operational for the ingestion pipeline and multi-document use cases.

### **Structured output with Pydantic schemas**

Every agent boundary is typed. RiskClause requires a verbatim `original_text` field. NegotiationSuggestion requires both `original_text` and `suggested_text`. This is not merely an engineering preference — it is a product guarantee.

### **Contract-type-aware calibration**

Clause does not apply one generic risk standard to every document. Each agent is prompted with contract-type-specific calibration rules, including different severity thresholds, different gap checklists, and explicit “do not flag” instructions for commercially standard provisions.

---

*Disclaimer: Clause is an AI-powered analysis tool and does not provide legal advice.*